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Smart Water Meter
To Help Control Water Wastage

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Increasing water consumption and wastage is leading to water scarcity. While a large population has no safe drinking water, some people use a lot of water on daily basis and waste it.

This project is of a smart water meter that measures our daily water usage and feeds live data on our phone that is connected to a database, which can be accessed by the concerned authorities through a web browser.

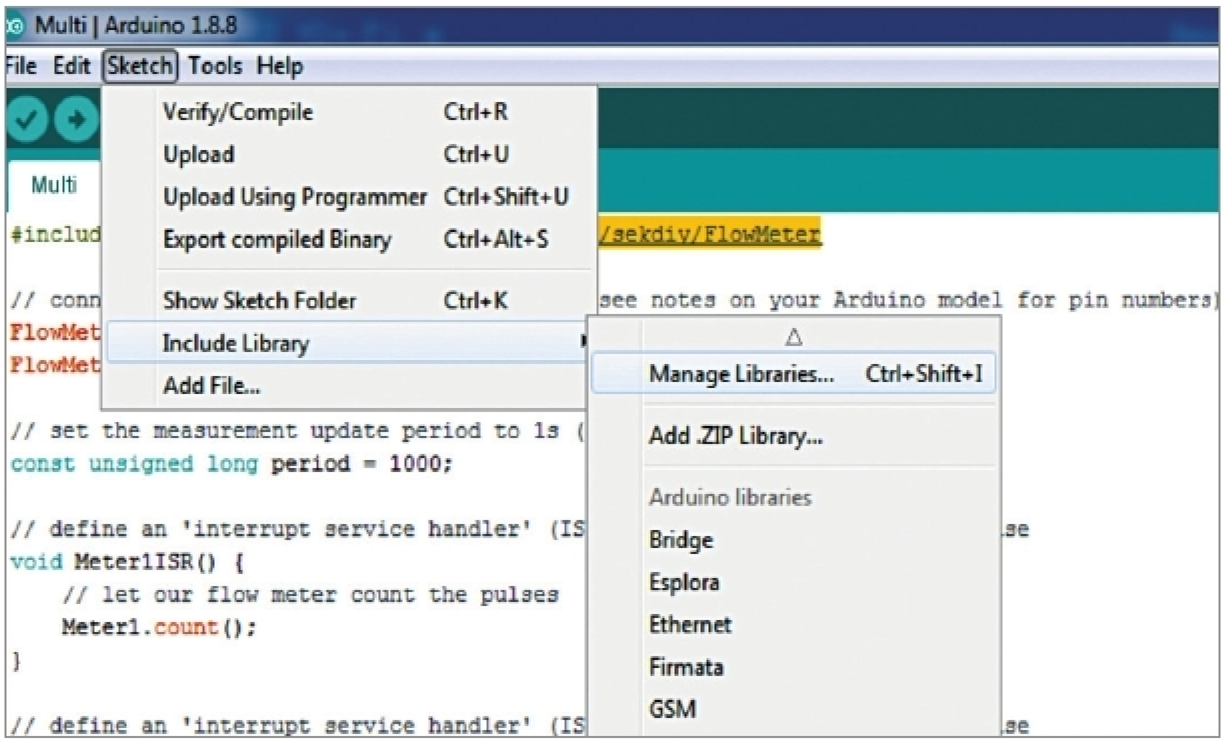

Fig. 1: Steps to add Arduino OLED library

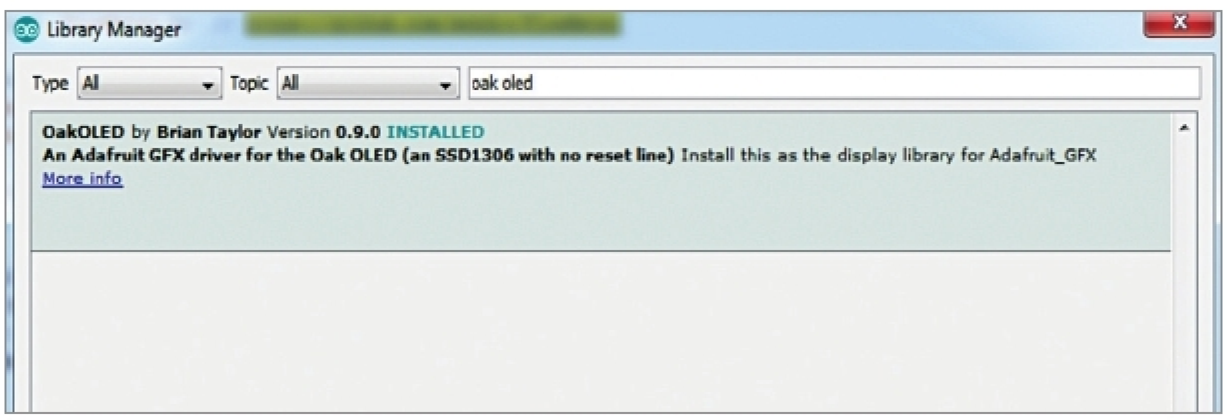

Fig. 2: Installing OLED library

```
#include <FlowMeter.h>  // https://github.com/sekdiy/FlowMeter

#include "Wire.h"
#include "OakOLED.h"
OakOLED oled;
FlowMeter Meter1 = FlowMeter(2);
FlowMeter Meter2 = FlowMeter(3);
```

Fig. 3: Arduino code with libraries

Some components required for the project are:

- Water flow sensor (FS400A is recommended).
- Arduino Pro Mini
- Bluetooth HC-05
- Programmer (Arduino Uno/ FTDI USB-UART adaptor)
- OLED SSD1306
- Water pipe (size as per flow meter inlet diameter)
- Water tap connector

To start the project, download the flow meter library from the link *https://github.com/sekdiy/FlowMeter*. Unzip the FlowMeter-master folder and paste it in the libraries folder of Arduino IDE. After that, open multi.ino file from FlowMeter-master folder and save it as water_meter.ino.

We need to include some additional features in this code by adding some libraries and modifying the code. To add libraries, click on Manage Libraries (refer Fig. 1) and search for OakOLED SSD 1306 library and install that library as shown in Fig. 2.

Now our Arduino IDE is ready. Let's start the coding part.

Coding

First, we need to include libraries in water_meter.ino code as shown in Fig. 3.

Next, create the interrupt han-

```
void Meter1ISR() {

   Meter1.count();
}


void Meter2ISR() {

   Meter2.count();
}
```

Fig. 4: Arduino Interrupt handler